

Haosui Duanmu

PROFESSOR

Institute for Advanced Study in Mathematics, Harbin Institute of Technology

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Education

University of Toronto

PHD STATISTICS

- Advisor: Prof. Jeffrey Rosenthal, Prof. William Weiss, and Prof. Daniel Roy

Toronto

2012 - 2017

University of Toronto

MS STATISTICS

- Advisor: Prof. Jeffrey Rosenthal and Prof. William Weiss

Toronto

2011 - 2012

University of Toronto

BS UNDERGRADUATE DEGREE IN MATHEMATICS

- Graduate with High Distinction
- Undergrad research advisor: Prof. Franklin Tall

Toronto

2007 - 2011

Employment

2022-
present

Professor, Institute for Advanced Study in Mathematics, Harbin Institute of Technology

2018-2022

Postdoctoral Fellow, Department of Economics, University of California, Berkeley

Publications

PUBLISHED

Robert M. Anderson and **Haosui Duanmu**. 2025. Equilibrium and Social Norm. *Games and Economic Behavior*, 154: 119-128.

Haosui Duanmu and Aaron Smith. 2025. Nonstandard representation of the Dirichlet form and applications to the comparison theorem. *Mathematische Nachrichten*, 298(4): 1167-1183.

Robert M. Anderson and **Haosui Duanmu**. 2025. Cap-and-trade and Carbon Tax Meet Arrow-Debreu. *Econometrica* (Lead Article), 93(2): 357-393.

Robert M. Anderson, **Haosui Duanmu** and Aaron Smith. 2024. Mixing times and hitting times for general Markov processes. *Israel Journal of Mathematics*, 259(2): 759-834.

Robert M. Anderson, **Haosui Duanmu**, M. Ali Khan and Aniruddha Ghosh. 2024. On existence of Berk-Nash equilibria in misspecified Markov decision processes with infinite spaces. *Journal of Economic Theory*, 217: 30 pages.

Haosui Duanmu, Daniel Roy and Aaron Smith. 2023. Existence of matching priors on compact spaces. *Biometrika*, 110(3): 763-776.

Haosui Duanmu, David Schrittemser and William Weiss. 2023. Loeb extension and Loeb equivalence II. *Fundamenta Mathematicae* 262(1): 71-83.

Robert M. Anderson, **Haosui Duanmu**, M. Ali Khan and Metin Uyanik. 2022. Walrasian equilibrium theory with and without free-disposal: theorems and counterexamples in an infinite-agent context. *Economic Theory* (Lead Article), 73(2-3): 387-412.

Robert M. Anderson, **Haosui Duanmu**, M. Ali Khan and Metin Uyanik. 2022. On abstract economies with an arbitrary set of players and action sets in locally-convex topological vector spaces. *Journal of Mathematical Economics*, 98: 9 pages.

Haosui Duanmu, Jeffrey Rosenthal and William Weiss. 2021. Ergodicity of Markov processes via nonstandard analysis. *Memoirs of the American Mathematical Society*, 273(1342): 114 pages.

Haosui Duanmu and Daniel Roy. 2021. On extended admissible procedures and their nonstandard Bayes risk. *The Annals of Statistics*, 49(4): 2053–2078.

Robert M. Anderson, **Haosui Duanmu** and Aaron Smith. 2021. Mixing and average mixing times for general Markov processes. *Canadian Mathematical Bulletin*, 64(3): 541–552.

Robert M. Anderson, **Haosui Duanmu** and Aaron Smith. 2021. Mixing and hitting times for Gibbs samplers and other non-Feller processes. *Illinois Journal of Mathematics*, 65(3): 547–577.

Robert M. Anderson, **Haosui Duanmu**, David Schritterser and William Weiss. 2021. Loeb extension and Loeb equivalence. *Proceedings of the American Mathematical Society, Series B*, 8: 112–120.

Haosui Duanmu and William Weiss. 2018. Finitely-additive, countably-additive and internal probability measures. *Commentationes Mathematicae Universitatis Carolinae*, 59(4): 467–485.

Haosui Duanmu, Franklin Tall and Lyubomyr Zdomskyy. 2013. Productively Lindelöf and indestructibly Lindelöf spaces. *Topology and its Applications*, 160(18): 2443–2453.

IN REVIEW

Robert Anderson, **Haosui Duanmu**, Ming Xiao and Yahui Zhang. 2025. Unstable Agents in Infinite Matching Markets. *Theoretical Economics*, Under review.

Robert Anderson and **Haosui Duanmu**. 2025. Equilibrium and social norm. *Games and Economic Behavior*, Revise and Resubmit.

IN PREPARATION

Robert Anderson and **Haosui Duanmu**. 2024. The Second Welfare Theorem for Quota Equilibrium. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5075138

Robert Anderson, **Haosui Duanmu**, M. Ali Khan and Metin Uyanik. 2024. Measure-theoretic Production Economies with Externalities and Bads. Draft available upon request.

Robert Anderson and **Haosui Duanmu**. 2025. Pareto Optimality with Externality. Draft available upon request.

Awards, Fellowships, & Grants_____

2019	NSERC Postdoctoral Fellowship , Natural Sciences and Engineering Research Council of Canada	90,000 CAD
2021	Overseas Excellent Young Scientists Fund Program , National Natural Science Fund	2 million RMB
2022	Young Scientist Fund Program , National Natural Science Fund	300,000 RMB

Teaching Experience_____

Winter 2017 **Nonstandard Analysis**, Instructor, *University of Toronto*

Winter 2024 **Calculus**, Instructor, *Harbin Institute of Technology*

Mentoring_____

2022-present **Tian Luo**, Master Student, Harbin Institute of Technology

2023-present **Yahui Zhang**, Phd Candidate, Harbin Institute of Technology

2023-present **Kelun Ma**, Master Student, Harbin Institute of Technology

2024-present **Xinyu Liu**, Phd Candidate, Harbin Institute of Technology